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Apparatus and methods for maintaining a gap

Abstract

A method and apparatus for maintaining a fixed distance between two cabinets during installation are provided. An elongated rail with two stops and two clamps are used. The stops are fixed to the elongated rail with opposing faces placed the fixed distance apart. The assembly is placed between the cabinets and a clamp is used to pinch a cabinet wall of each cabinet against a stop. The cabinets are installed and the apparatus is removed. In some embodiments a second apparatus may be used, one at the top of the cabinets and one at the bottom.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a division of U.S. patent application Ser. No. 17/547,220, filed Dec. 10, 2021, which claims the benefit of U.S. Provisional Patent Application No. 63/123,316, filed Dec. 9, 2020, both of which are hereby incorporated by reference in their entirety.

FIELD

(1) The present invention generally relates to products and apparatus for maintaining a spacing between two or more objects. In particular, the apparatus and methods may be used for maintaining a spacing during the installation of cabinets.

BACKGROUND

(2) During the installation of cabinets, such as in a kitchen installation, it is often necessary to maintain a consistent and precise space between two cabinets that are being installed. For example, many appliances such as stoves and dishwashers have standard widths and cabinets need to be installed on either side of a gap with the correct width to allow the appliance to be installed later.

(3) While it would seem that maintaining such a width would be easily accomplished by simply measuring, this is not the case. During the cabinet installation process, the cabinets are placed into position and then must be leveled. During the leveling process, the cabinets can encroach into the gap and consequently render the gap too small for the later installation of the appliance.

(4) This effect is exacerbated by the close tolerances that may be desired. Large gaps between the appliances and the cabinets are not only unsightly, but also create unwanted crumb and dirt traps after installation is complete. Accordingly, it is very desirable to maintain any spacing between cabinets designed for an appliance as accurately as possible.

SUMMARY OF THE EMBODIMENTS

(5) Objects of the present patent document are to provide improved methods and apparatus for maintaining a gap between cabinets during and installation. While the apparatus can be used for maintaining a gap between any two objects, its preferred use is to maintain a gap between cabinets during the installation of the cabinets, for example in a kitchen.

(6) A method for maintaining a fixed distance between two cabinets during installation is provided. In a preferred embodiment, the method comprises adjusting a first stop along an axial length of an elongated rail. Then, fixing the first stop in a first position such that a first flat surface of the first stop is perpendicular to the axial length. Adjusting a second stop along the axial length of the elongated rail. Then, fixing the second stop in a second position such that a second flat surface of the second stop is perpendicular to the axial length and is parallel and facing an opposite direction from the first flat surface, wherein the first flat surface and the second flat surface are the fixed distance apart. Once the stops are positioned and fixed to the elongated rail, the elongated rail with the first stop and second stop is placed between the two cabinets such that the first surface interfaces with a first cabinet of the two cabinets and the second flat surface interfaces with a second cabinet of the two cabinets. Then, a first clamp is adjusted along the axial length of the elongated rail. Once the first clamp is positioned, it may be fixed in a third position adjacent the first stop such that a first wall of the first cabinet is between the first stop and the first clamp. The first clamp is tightened to squeeze the first wall between the first stop and the first clamp. Next, a second clamp is adjusted along the axial length of the elongated rail. Once the second clamp is in place, it is fixed in a fourth position adjacent the second stop such that a second wall of the second cabinet is between the second stop and the second clamp. The second clamp is tightened to squeeze the second wall between the second stop and the second clamp. Once everything is secured, the first cabinet and second cabinet can be fixed in place by screwing or gluing or some other method or any combination thereof. Finally, once the cabinets are installed and fixed in place, the first clamp and the second clamp can be loosened and the elongated rail with the first stop, second stop, first clamp and second clamp removed from between the two cabinets.

(7) In some embodiments, the elongated rail has a channel that runs along the axial length. In some embodiments, the cross-section of the channel is dovetail shaped.

(8) In general, the stops and clamps may be secured to the rail using many different methods. In one embodiment, the first stop, second stop, first clamp and second clamp are all fixed to the rail by tightening a fastener with a flange coupled to the fastener within the channel. In some embodiments, the flange may be a t-nut.

(9) In preferred embodiments, the first stop and second stop each have ridges that extend into the channel. The ridges preferably fill most of the gap between the channel in order to help keep the stops perpendicular to the axial length of the rail and not allow the stops to rotate with respect to the axial length of the rail. Even more preferably, the first stop and second stop each have two ridges that extend into the channel on either side of a fastener.

(10) In preferred embodiments, the first clamp and second clamp each have a tip that is designed to squeeze an object against a stop and is adjustable along an axial length of the first clamp and second clamp via a thread.

(11) In some embodiments, the first clamp and second clamp each have two fasteners and two flanges that couple the first clamp and the second clamp to the elongated rail respectively.

(12) In preferred embodiments, the elongated rail has preset positions along the axial length for the first stop and the second stop to allow the first stop and second stop to be positioned the fixed distance apart. The preset positions may be at 30 inches, 36 inches and 48 inches to name a few.

(13) In another aspect of the embodiments taught herein, an apparatus for maintaining a fixed distance between two objects is provided. In preferred embodiments, the apparatus comprises an elongated rail with an axial length and a channel running along the axial length. A first stop releasably coupled to the elongated rail such that the first stop is adjustable along the axial length of the elongated rail and fixable in various positions along the axial length of the elongated rail.

The first stop has a first flat surface that is perpendicular to the axial length when the first stop is in a first fixed position along the axial length of the elongated rail. Preferably, the first stop has a first ridge that extends into the channel. A second stop releasably coupled to the elongated rail such that the second stop is adjustable along the axial length of the elongated rail and fixable in various positions along the axial length of the elongated rail. The second stop has a second flat surface that is perpendicular to the axial length and facing in an opposite direction from the first flat surface when the first stop is in a second fixed position along the axial length of the elongated rail. Preferably, the second stop has a second ridge that extends into the channel. A first clamp that is releasably coupled to the elongated rail such that the first clamp is adjustable along the axial length of the elongated rail and fixable in various positions along the axial length of the elongated rail. The first clamp has a first tip designed to press an object against the first stop. Preferably, the first tip is adjustable along the axial length of the first clamp. A second clamp releasably coupled to the elongated rail such that the second clamp is adjustable along the axial length of the elongated rail and fixable in various positions along the axial length of the elongated rail. The second clamp has a second tip designed to press an object against the second stop. Preferably, the second tip is adjustable along the axial length of the second clamp.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an isometric view of one side of one embodiment of an apparatus for maintaining a fixed distance between two objects.
- (2) FIG. 2 illustrates a cross-sectional view of one embodiment of a rail including a dove tail shaped channel for use as part of the apparatus for maintaining a fixed distance between two objects.
- (3) FIG. 3 illustrates an isometric view of one embodiment of a stop for use in an apparatus for maintaining a fixed distance between two objects.
- (4) FIG. 4 illustrates an isometric view of a second embodiment of a stop for use in an apparatus for maintaining a fixed distance between two objects
- (5) FIG. 5 illustrates a plan view of another embodiment of a stop **14** mounted to a rail **12** for use in an apparatus for maintaining a fixed distance between two objects.
- (6) FIG. 6 illustrates one embodiment of a clamp for use in an apparatus for maintaining a fixed distance between two objects.
- (7) FIG. 7 illustrates a side view of the clamp of FIG. 6.
- (8) FIG. 8 illustrates an isometric view of one end of an apparatus for maintaining a fixed distance between two objects with the stop and clamp coupled to the rail.
- (9) FIG. 9 illustrates a view of the plate **16** without the clamping mechanism attached.
- (10) FIG. 10 illustrates the plate of FIG. 9 with nothing attached to the plate.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (11) The present patent application discloses an apparatus, device or tool that maintains a fixed and precise distance between two material bodies during an installation or fabrication process. The distance is maintained by the device even during a leveling process of the material bodies or other movements that may be required for installation of the bodies.
- (12) The device maintains the precise fixed distance by clamping on to the materials and maintains the fixed dimension after clamps are engaged. The device accommodates for a range of material thickness and distances.
- (13) By utilizing rods, clamps and stops in concert with one another, the device maintains the desired fixed dimension of space between bodies. In addition to the precise spacing it maintains, the device also keeps material bodies parallel and plumb.

(14) The device consists of a system of components that includes various rod lengths and can be packaged as a system of stops and clamps.

(15) Components

(16) FIG. 1 illustrates an isometric view of one side of one embodiment of an apparatus for maintaining a fixed distance between two objects **10**. As may be appreciated, the apparatus **10** is coupled to a cabinet in FIG. 1. In some embodiments, the apparatus **10** is coupled between two cabinets or stated another way, coupled to two cabinets, one on each end of the apparatus **10**.

(17) As may be seen in FIG. 1, the apparatus **10** includes a rail **12**. The rail **12** is an elongated body with an axial length extending along its elongated body. The rail **12** can be manufactured in any length. Common lengths of gaps that need to be maintained in the cabinet industry are 30", 36", 48", and 60". The length of the rail **12** needs to be longer than the desired opening size or gap that needs to be maintained. To this end, the axial length of the rail **12** is preferably thirty inches or more. The rail **12** can be made of any ridged material such as, aluminum, wood, steel, copper, cobalt, plastic, ceramic, iron, cast iron. Preferably, the rail **12** is made from an extruded aluminum or steel.

(18) In preferred embodiments, rail **12** has a flat surface **29** along the entire axial length. It will be understood from the teachings below, flat surface **29** is important to allow stops to be coupled to the rail **12** with surfaces perpendicular to the axial length of the rail **12**. However, in other embodiments the rail **12** could be cylindrical and the stops could be coupled around the rail and fixed with a set screw or other means of securing.

(19) FIG. 2 illustrates a cross-sectional view of a preferred rail **12** showing a dove tail type channel **15** running the entire length of the rail **12**. Although not required, preferably the rail has a long channel, slot or groove **15** running the entire length. In some embodiments, the channel **15** may not run the entire length but only a portion of the length. In embodiments where the channel **15** does not run the entire length of the rail **12**, the channel **15** is preferably found at each distal end of the rail **12** and absent in the middle of the rail **12**.

(20) Preferably, the channel **15** has a dove tail shape but other shapes are possible such as rectangular or U-shaped or other shapes. The key to the channel shape is that it works to help capture the components of the apparatus as they are adjusted, and preferably slidably adjusted, along the length of the rail **12**. To this end, regardless of the ultimate shape of the channel **15**, it preferably has lips **11** that help form a channel that can retain any slidable components. In other embodiments, the rail **12** has no channel **15** and the components may be releasably coupled to the exterior of the rail **12**.

(21) FIG. 3 illustrates an isometric view of one embodiment of a stop **14** for use in an apparatus for maintaining a fixed distance between two objects. Stop **14** can be any device that may be coupled to rail **12** and be locked into a particular position at different distances along the axial length of the rail **12**. In preferred embodiment, stop **14** is slidably coupled to rail **12** and more preferably, slidably coupled to the channel **15** in the rail **12**. In embodiments with a channel **15**, the stop is preferably captured by the channel **15**. By captured it is meant that the stop **14** can be loosened and slid along the axial length of the rail **12** but it will not easily separate from the rail **12**. In embodiments that don't include a slot or channel **15**, the stop may be coupled to an outside surface of the rail **12**. Importantly, the stop **14** needs to be coupled in a way that is lockable in different positions along the axial length of the rail **12**.

(22) In the embodiment shown in FIG. 3, the stop **14** is generally blocked shaped. However, stops **14** can be any other shape but should have at least one flat surface **13**. Preferably stop **14** is made from a ridged material such as metal, preferably aluminum or steel. However, any metal may be used and plastics, ceramics and other rigid materials may also be used without departing from the scope of the embodiments. Ideally, the stop is rigid so it cannot be compressed. This is important for maintaining the exact distance between the objects desired. However, in some embodiments, a thin compressible layer of material may be added to surface **13** to protect the objects stops **14** will

interface to. This thin compressible material may be rubber or a similar material. The compressible nature of this material should be within the allowable tolerances of the gap that needs to be maintained such that in operation the compression of the material does not modify the desired gap distance.

(23) The stop is as accompanied by an adjustable fastener **24**. In the embodiment shown in FIG. 3, the adjustable fastener **24** is a threaded fastener. However, in other embodiments, the adjustable fastener **24** may be a clip, spring, spring lock, slide-lock, screw or any other type of fastener that can be adjusted to releasably couple the stop **14** to the rail **12**.

(24) In some embodiments, the stop **14** may also have a capture flange **17**. In the embodiment shown in FIG. 3, the capture flange **17** is a T-nut **17**. However, any type of flange may be used. Preferably, the capture flange **17** is coupled to the adjustable fastener **17** such that it can be tightened and loosened with respect to the body of the stop **14**. The capture flange **17** is designed to fit within the channel **15** behind lips **11**. This allows the stop to be captured in the channel **15** and to the rail **12** but slidably connected to the rail **12**. Stops **14** can be made from a solid block of material or L-shaped bracket. A stop **14** has a surface **13** that when the stop **14** is fixed to the rail **12**, surface **13** is perpendicular to the axial length of the rail **12**.

(25) Stops **14** can be slid in either direction in the groove **15** of the rail **12** for desired placement. The apparatus **10** has two stops, one at each distal end. Both stops **14** are adjusted to the desired opening, then tightened down (secured) using a fastener **24** and capture flange **17**. In the embodiment shown in FIG. 3, a threaded fastener such as a cap screw is used in combination with a T-Nut **17**. The threaded fastener **24** passes through a hole in the stop **14**. The flange **17** is placed under the lips **11** of the channel **15** and when the stop **14** is located the correct distance along the rail **12**, the screw **24** is tightened with a tool such as an allen wrench or similar to lock the stop **14** into the desired position along the axial length of the rail **12**. At this stage, surface **13** is perpendicular to the axial length of the rail **12**.

(26) In some embodiments, stop **14** has a threaded insert to allow fastener **24** to thread into the stop **14**.

(27) In addition, the surface of stop **13** that interfaces with rail **12** is perpendicular to surface **13**. In some embodiments, stop **14** has a pair of ridges **22**. Ridges **22** extend down from the stop **14** and are designed to extend into channel **15**. Preferably, ridges **22** are located on either side of the fastener **24** on a surface perpendicular to surface **13**. Ridges **22** ensure that surface **13** is perpendicular to the axial length of the rail **12** when the stop is locked into position along the rail **12**.

(28) FIG. 4 illustrates an isometric view of a second embodiment of a stop **14** for use in an apparatus for maintaining a fixed distance between two objects. The stop **14** in FIG. 4 is similar and works in a similar manner to the stop **14** in FIG. 3. The stop if FIG. 4 is shaped in the shape of a cube with ridges **22** protruding from the surface designed to interface with the rail **12**. In FIG. 4, the capture flange **17** is a square nut.

(29) FIG. 5 illustrates a plan view of another embodiment of a stop **14** mounted to a rail **12** for use in an apparatus for maintaining a fixed distance between two objects. The stop **14** in FIG. 5 is similar and works in a similar manner to the stop **14** in FIG. 4. The stop if FIG. 5, has a capture flange **17** that is shaped to fit in the dove tail channel **15**. In particular, the capture flange **17** has a cross-section that allows a portion of the capture flange **17** to fit up in between lips **11** of the channel **15**. Although the embodiments of stops **14** have been shown with a single fastener **24** and capture flange **17**, in other embodiments the stops **14** may have two, three or more fastener/capture flange combinations.

(30) In operation, at least two stops **14** are used and slidably coupled to the rail **12**. The stops are mounted to the rail **12** with their flat surfaces **13** facing away from each other. The stops **14** are adjusted along the rail **12** until the distance between the outside edges or surfaces **13** of each stop **14** defines the gap to be maintained. The stops **14** are then secured to the rail **12** such that they

cannot move with respect to each other or the rail **12**. This is preferably done by tightening the fastener **24** of each stop **14**.

(31) In preferred embodiments, the rail **12** may be manufactured with preset positions for the stops at particularly common distances. To this end, there may be notches or other changes in the rail that allow the stops to easily be located such that their surfaces **13** are at exactly 30 inches apart or other commonly used distances. As just one example, the rail **12** may have small holes or dents and the stops **14** may be manufactured with a spring captured ball that seats into the holes in the rail **12** to allow predetermined spacing of the stops **14**. Markings could be added to the rail **12** to allow visual confirmation of the spacing of the stops at any predetermined locations.

(32) FIG. **6** illustrates one embodiment of a clamp for use in an apparatus for maintaining a fixed distance between two objects. In the embodiment shown in FIG. **6**, the clamp includes a plate **16**. Plate **16** is used to couple the clamp **18** to the rail **12** and in other embodiments other coupling methods may be used. The plate **16** is made from a ridged material and the clamping device is coupled to the plate **16** by either fasteners **24** or permanently such as in welding or construction from a single piece.

(33) Similar to the stop **14**, the clamp **18** needs to be able to be locked into particular locations along the axial length of the rail **12** and is in preferred embodiments captured to the rail and slidably coupled to the rail **12**. The clamp **18** moves along the longitudinal axis of the rail **12** and may be fixed in place anywhere along the longitudinal axis of the rail **12**. In preferred embodiments, the clamp **18** can slide along the rail **12** using the groove or channel **15** to couple the clamp assembly to the rail **12** just like the stop **14**.

(34) FIG. **7** illustrates a side view of the clamp **18** of FIG. **6**. Clamp **18** may be designed in many different ways. In the embodiment shown in FIG. **7**, the clamp **18** includes a plunger device **21** mounted parallel to the plate **16**. The plunger **21** is coupled to a lever (handle) **25** at the distal end of the plunger **21**. At the opposite end of the plunger **21** from the lever **25** is a tip **19**.

(35) As may be appreciated in FIG. **7**, a bracket **29** is secured to the plate **16** with a fastener. The bracket **29** is an L shaped bracket with a hole that holds the plunger **21** in a parallel position to the plate **16**. The bracket **29** has sleeve **23** to ensure the plunger **21** remains parallel to the plate **16**.

(36) In operation, the plunger **21** can traverse back and forth along the longitudinal length of the plate **16** by operation of the handle **25**, which is attached to the distal end of the plunger **21**. In the embodiment shown in FIG. **7**, the plunger handle **25** is attached with a pivot **31** over a fulcrum such that when the handle **25** is pressed down, the plunger **21** is fixed in place.

(37) In the embodiment shown in FIG. **7**, the interface between the tip **19** and the plunger **21** is threaded. This allows for a fine adjustment of the tip **19** even when the clamp **18** is fixedly locked in position on the rail **12**.

(38) The distal end of the plunger **21** opposite the handle **25** has the tip. The tip may be made from a material that compresses slightly such as a rubber. In preferred embodiments, the tip **19** is made from a firm rubber. However, in other embodiments it may be made from other materials. Unlike the stops which have to maintain a fixed distance, the clamps **18** just lock the apparatus **10** together such that the tip **19** does not have the same tolerance requirements for the compression of the tip discussed in relation to the stop **14**. To this end, the tip **19** may be made of a softer material that compresses as long as it can hold the apparatus **10** in place on the cabinet or other object.

(39) The plunger **21** handle **25** and tip **19** may be part of a straight line toggle clamp or other type of clamp. The clamp is coupled to the plate **16** and the plate **16** interfaces with the rail **12**. As may be appreciated, there are a pair of fasteners **24** and capture flanges **17** that go through holes in the plate **16** and allow the clamp **18** to be captured to the rail **12**, slidably adjusted and locked into position in the same way the stop **14** is. In particular, once in the desired location, the user can turn the screw **24** to reduce the gap between the capture flange **17** and the under side of the channel **15** to fix the clamp **18** in place. Of course, in this embodiment the clamp has a pair of fasteners/capture flanges and the stop only has one. However, in other embodiments the clamp **18** may have only one

screw/capture flange or it may have more than two.

(40) The screws **24** and capture flanges **24** pass through one or more holes in the plate **16**. As discussed, this allows the clamp **18** to be slidably coupled to the rail **12** but also fixed in a single position.

(41) FIG. **8** illustrates an isometric view of one end of an apparatus **10** for maintaining a fixed distance between two objects with the stop **14** and clamp **18** coupled to the rail **12**. In operation, the stop **14**, along with the second stop on the opposite distal end of the rail, are distanced apart such that their surfaces **13** are the exact distance of the gap that is required to be maintained between objects. Once the correct distance apart, the stops **14** are locked in place by turning fastener **24**. The apparatus **10** is then slid between the objects such that each surface **13** of one of the stops **14** interfaces with one of the objects and separates them by the desired distance. The clamp **18** is then translated along the axial length of the rail **12** such that it is in position to firmly hold to the object when the handle **25** of the clamp is depressed and the tip **19** is translated along the axial length of the rail towards the stop **14**. Once the clamp **18** is in position it is fixed or locked down to the rail by turning screws **24** on the clamp **18**. Ideally, the same type of fasteners or at least fastener interfaces are used on all fasteners such that a single tool can be used for tightening and loosening the fasteners **24**. The same steps are repeated on the opposite distal end of the apparatus with the second stop and second clamp. In preferred embodiments, more than one apparatus may also be used such that two or more apparatus are placed between objects where it is desired to maintain a fixed distance between the objects. In the preferred embodiments, two apparatus are used, one near the top of the objects and one near the bottom. In other embodiments, a single apparatus is used in the middle of the objects or at the top or bottom of the objects.

(42) As discussed, coupling the tip **19** to the plunger **21** with threads allows fine adjustment of the position of the tip **19** with respect to the clamp **18** and, thus, the amount of force applied through the object to the stop **14** when the clamp **18** is locked down with the handle **25**. The clamp **18** can be adjusted to the desired pressure on the object, depending on the texture and sensitivity of the material, by adjusting the tip **19** via the threaded coupling. The clamp **18** is released by simply pulling the handle **25** up and allowing the plunger **21** to travel parallel to the plate **16** away from the stop **14**.

(43) As may be appreciated in FIG. **8**, when the lever (handle) **25** is utilized, the plunger **21** translates parallel to the plate **16** which corresponds to along the axial length of the rail **12**. The translation causes the tip **19** to engage securely against the object that is trying to be located and holds the apparatus to the object by squeezing a side of the object between the stop **14** and the clamp **18**.

(44) FIG. **9** illustrates a view of the plate **16** without the clamping mechanism attached. As may be appreciated, the plate **16** can include a plurality of holes **35**. The plurality of holes allow the clamping mechanism to be coupled to the plate in various different locations. While as part of typical use the clamping mechanism will not need adjusted with respect to the plate **16**, the additional holes **35** allow for overall adjustment. In preferred embodiments, the holes **35** are predrilled threaded holes.

(45) As may also be seen in FIG. **9**, cap screws may be used as the fasteners **24** that attach the clamp **18** to the rail **12** in combination with the capture flange **17**.

(46) FIG. **10** illustrates the plate **16** of FIG. **9** with nothing attached to it.

(47) Similar to the stops **14**, the apparatus **10** includes a first and second clamp **18** each at opposite distal ends of the rail **12** with their tips pointing towards the middle and towards the respective stops **14**.

(48) As may be appreciated, the clamps **18** are used to secure the device **10** into the gap by squeezing a side wall or other portion of the cabinet or other object between the clamp **18** and the stop **14**. As may be seen in FIG. **1**, a stop **14** is on one side of the cabinet and the clamp **18** on the other. The clamp **18** applies pressure through the cabinet to the stop **14**. This is repeated on the

other side, opposite distal end, of the gap with an additional stop **14** and clamp **18** coupled to the same rail **12** on the opposite distal end, and thus the system is secured into the gap. Once attached on both sides and secured, the apparatus **10** will ensure the exact gap is maintained between the two objects.

(49) In use, the components of the apparatus **10** may be coupled to the rail **12** and are preferably slidably coupled to the rail **12**. In addition, other elements such as levels or other components may be coupled to the rail **12**. The groove **15** allows both the stop **14** and clamp **18** to be securely positioned anywhere along the rail **12**.

(50) Motivating Challenge Addressed

(51) As just one example of the device's use, the device may be used to maintain a fixed dimension of separation between two cabinets to accommodate an appliance such as a range or dishwasher. The precise opening needs to be maintained while leveling cabinets on both sides. In the process of leveling cabinets, the gap between the cabinets may often change. The apparatus described herein prevents a change in the dimensions of the gap and thus, allows easier installation of objects such as cabinets.

(52) In the case of kitchen cabinet installation, the dimensions of the opening are determined by the width of the appliance which becomes the fixed dimension. Both cabinets on the left and right need to be maintained at a precise distance from one another and stay parallel in order to later accommodate the appliance. Once the cabinets are levelled attached to the wall, or other anchor point, the device can be removed.

(53) Some Example Use Cases

(54) Though there are many possible uses for this device, following are two simple use cases.

(55) Use Case 1

(56) A cabinet installer would use the device for appliance openings such as refrigerator, washer/dryer, dishwasher, range, microwave, or a hood vent. The Device/tool can be left engaged for any desired amount of time. The device would be removed once appliances are ready to be installed or removed after counter tops are installed.

(57) In a preferred embodiment, the system would consist of 1 rail, two stops and two clamps. More than one system may be used at a time for a single gap and extra stops and clamps may be provided.

(58) In operation, the installer would determine the size of the gap needed between two cabinets. The stops would be slid manually along the longitudinal length of the rail and secured in place on opposite distal ends at the exact spacing the installer wishes to maintain. One end of the device is then secured to a first cabinet by adjusting a first clamp along the longitudinal length of the rail to squeeze the cabinet wall between the first clamp and the stop. A second clamp is used on the opposite end to secure the device onto the second cabinet on the other side of the gap, thus maintaining the gap. In some embodiments, the steps are repeated with a second device.

(59) Use Case 2

(60) In addition, to cabinet installers, the device could be used for numerous other cases where a consistent gap needs to be maintained. For example, a welder could benefit from use of this device while welding a rectangle. Both material bodies must be held a fixed distance and parallel to each other. Once clamps and stops are positioned and engaged, this allows the welder to fabricate other sides. Once fabrication is complete, the device can be removed.

Claims

1. A method for maintaining a fixed distance between two cabinets during installation comprising: adjusting a first stop along an axial length of an elongated rail with a channel that runs along the axial length; fixing the first stop in a first position by tightening a first fastener coupled to a first flange within the channel such that a first flat surface of the first stop is perpendicular to the axial

length; adjusting a second stop along the axial length of the elongated rail; fixing the second stop in a second position by tightening a second fastener coupled to a second flange within the channel such that a second flat surface of the second stop is perpendicular to the axial length and is parallel and facing an opposite direction from the first flat surface, wherein the first flat surface and the second flat surface are the fixed distance apart; placing the elongated rail with the first stop and second stop between the two cabinets such that the first surface interfaces with a first cabinet of the two cabinets and the second flat surface interfaces with a second cabinet of the two cabinets; adjusting a first clamp along the axial length of the elongated rail; fixing the first clamp in a third position adjacent the first stop by tightening a third fastener coupled to a third flange within the channel such that a first wall of the first cabinet is between the first stop and the first clamp; tightening the first clamp to squeeze the first wall between the first stop and the first clamp; adjusting a second clamp along the axial length of the elongated rail; fixing the second clamp in a fourth position adjacent the second stop by tightening a fourth fastener coupled to a fourth flange within the channel such that a second wall of the second cabinet is between the second stop and the second clamp; tightening the second clamp to squeeze the second wall between the second stop and the second clamp; fixing the first cabinet and second cabinet in place; and loosening the first clamp and the second clamp and removing the elongated rail with the first stop, second stop, first clamp and second clamp from between the two cabinets.

2. The method of claim 1, wherein a cross-section of the channel is dovetail shaped.

3. The method of claim 1, wherein the first stop and second stop each have ridges that extend into the channel.

4. The method of claim 1, wherein the first stop and second stop each have two ridges that extend into the channel on either side of a fastener.

5. The method of claim 1, wherein the first clamp and second clamp each have a tip that is designed to squeeze an object against a stop and is adjustable along an axial length of the first clamp and second clamp via a thread.

6. The method of claim 1, wherein the first flange, second flange, third flange and fourth flange are a t-nut.

7. The method of claim 1, wherein the elongated rail has preset positions along the axial length for the first stop and the second stop to allow the first stop and second stop to be positioned the fixed distance apart.

8. An apparatus for maintaining a fixed distance between two cabinets during installation comprising: an elongated rail with an axial length; a first stop adjustably coupled to the elongated rail such that the first stop can translate along the axial length of the elongated rail; wherein the first stop has a first flat surface perpendicular to the axial length of the elongated rail; a first fastener designed to fix a first position of the first stop with respect to the elongated rail when the first fastener is engaged; a second stop adjustably coupled to the elongated rail such that the second stop can translate along the axial length of the elongated rail; wherein the second stop has a second flat surface perpendicular to the axial length of the elongated rail; a second fastener designed to fix a second position of the second stop with respect to the elongated rail when the second fastener is engaged such that the first flat surface and second flat surface define a predetermined distance apart; a first clamp adjustably coupled to the elongated rail such that the first clamp can translate along the axial length of the elongated rail; a third fastener designed to fix a third position of the first clamp with respect to the elongated rail when the third fastener is engaged such that a first wall of a first cabinet is squeezed between the first flat surface and the first clamp; a second clamp adjustably coupled to the elongated rail such that the second clamp can translate along the axial length of the elongated rail; and a fourth fastener designed to fix a fourth position of the second clamp with respect to the elongated rail when the fourth fastener is engaged such that a second wall of a second cabinet is squeezed between the second flat surface and the second clamp such that the first cabinet and second cabinet are the predetermined distance apart.

9. The apparatus of claim 8, wherein the elongated rail has a channel that runs along the axial length.
10. The apparatus of claim 9, wherein a cross-section of the channel is dovetail shaped.
11. The apparatus of claim 9, wherein the first fastener, second fastener, third fastener and fourth fastener all comprise a flange within the channel.
12. The apparatus of claim 9, wherein the first stop and second stop each have ridges that extend into the channel.
13. The apparatus of claim 12, wherein the first stop comprises a first two ridges that extend into the channel on either side of the first fastener and the second stop comprises a second two ridges that extend into the channel on either side of the second fastener.
14. The apparatus of claim 8, wherein the first clamp and second clamp each have a tip that is designed to squeeze an object against a stop and is adjustable along an axial length of the first clamp and second clamp via a thread.
15. The apparatus of claim 11, wherein the flange is a t-nut.
16. The apparatus of claim 9, wherein the first clamp and second clamp each have two fasteners and two flanges that couple the first clamp and the second clamp to the elongated rail respectively.
17. The apparatus of claim 8, wherein the elongated rail has preset positions along the axial length for the first stop and the second stop to allow the first stop and second stop to be positioned the predetermined distance apart.
18. The apparatus of claim 8, wherein the first clamp and second clamp each comprise a plunger coupled to a lever designed to translate the plunger along the axial length of the elongated rail.
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